AI-BASED FINANCIAL NEWS SENTIMENT ANALYSIS FOR STOCK MARKET PREDICTION

A Mini Project Report Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

By

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Under the Esteemed Guidance of

Dr.K Swanthana



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K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

The mini-project report entitled "AI-Based Financial News Sentiment Analysis for Stock Market Prediction" is a record of bona fide work carried out by Roopesh (2420030359), Deepthi Anusha (2420030448), and Vyshnavi (2420030449) in partial fulfillment of the requirements for the award of the B.Tech degree at K L (Deemed to be) University. The work presented in this report is original to the best of our knowledge, and all datasets, tools, code, and external sources used have been appropriately acknowledged.

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CERTIFICATE

This is to certify that the mini-project report entitled “AI-Based Financial News Sentiment Analysis for Stock Market Prediction” is a bona fide work carried out by Roopesh (2420030359), Deepthi Anusha (2420030448), and Vyshnavi (2420030449) under the supervision of Dr. K. Swanthana during the academic year 2026–2027, in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

Signature of the Project Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide, Dr. K. Swanthana, for her valuable guidance, encouragement, and support throughout this project. Her suggestions helped us improve our understanding and implementation of the project.
We thank K L (Deemed to be) University and the Department of Computer Science and Engineering for providing the facilities and resources required to complete this mini-project.
We also thank our faculty members, friends, and families for their continuous support and encouragement during the completion of this work.

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We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

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1. ABSTRACT

Stock-market movements are influenced by financial news, investor sentiment, and historical market trends. This project presents an AI-based approach to predicting stock-market direction by combining financial news sentiment with historical stock data. The TRACE ACL18 AAPL dataset, containing 504 daily records, is used for the analysis. FinBERT, a pretrained financial language model, extracts positive, negative, and neutral sentiment scores from financial text. These scores are combined with 35 engineered features covering stock returns, momentum, volatility, RSI, MACD, Bollinger Bands, and trading volume. The proposed FinSent-Momentum Fusion (FSMF) framework uses a Gradient Boosting classifier with 150 estimators, a learning rate of 0.05, and a maximum depth of 2. Logistic Regression and an adapted two-layer LSTM model are also evaluated for comparison. The FSMF model achieves 69.12% test accuracy and a 71.23% F1-score. A Streamlit interface is developed to demonstrate the prediction workflow. The project shows how financial text analysis and market indicators can be combined to support stock-market trend analysis.

Team Members and Responsibilities:

Roopesh – 2420030359: Model development
Deepthi Anusha – 2420030448: Data preprocessing
Vyshnavi – 2420030449: Results and documentation

2. INTRODUCTION

2.1 Background

Financial news plays an important role in influencing stock-market movements. News about company performance, earnings, economic conditions, and investor confidence can affect stock prices. Analysing this information manually is difficult because large amounts of financial text are published every day. Natural Language Processing (NLP) and machine learning can automatically analyse financial text and identify useful sentiment patterns.

2.2 Motivation

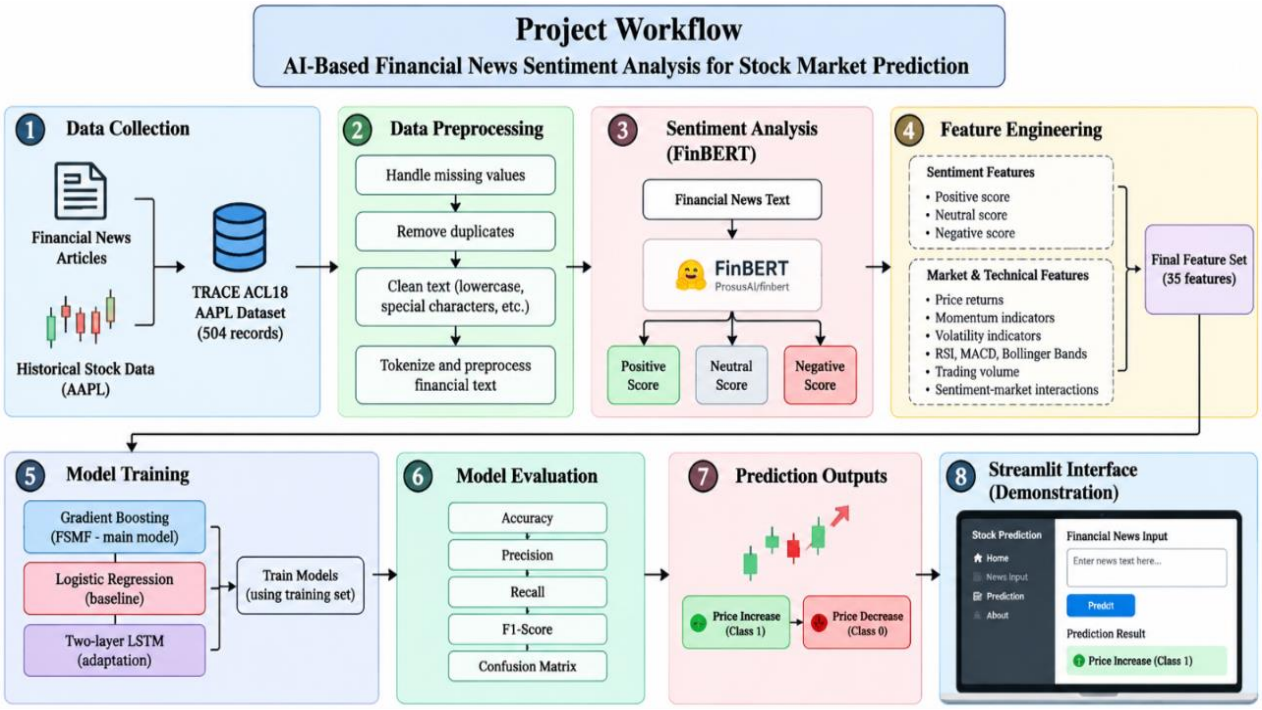
Traditional stock prediction approaches mainly depend on historical prices and technical indicators. However, financial news also contains information that may influence investor decisions. This project combines financial news sentiment with historical market data to improve the analysis of stock-market direction.

2.3 Proposed System

The proposed system uses FinBERT to extract sentiment scores from financial news. These scores are combined with technical indicators and historical stock features using the FinSent-Momentum Fusion (FSMF) framework. A Gradient Boosting classifier predicts stock direction, while Logistic Regression and an adapted two-layer LSTM are used for comparison. A Streamlit application demonstrates the prediction process.

2.4 Intended Users

The system is intended for students, researchers, and people interested in financial data analysis. It can help users understand how financial news sentiment and market indicators can be used together for stock-market prediction.


Project Workflow
AI-Based Financial News Sentiment Analysis for Stock Market Prediction
1 Data Collection
Financial News Articles
Historical Stock Data (AAPL)
TRACE ACL18 AAPL Dataset (504 records)
2 Data Preprocessing
Handle missing values
Remove duplicates
Clean text (lowercase, special characters, etc.)
Tokenize and preprocess financial text
3 Sentiment Analysis (FinBERT)
Financial News Text
FinBERT
ProsusAI/finbert
Positive Score
Neutral Score
Negative Score
4 Feature Engineering
Sentiment Features
• Positive score
• Neutral score
• Negative score
Market & Technical Features
• Price returns
• Momentum indicators
• Volatility indicators
• RSI, MACD, Bollinger Bands
• Trading volume
• Sentiment-market interactions
Final Feature Set (35 features)
5 Model Training
Gradient Boosting (FSMF - main model)
Logistic Regression (baseline)
Two-layer LSTM (adaptation)
Train Models (using training set)
6 Model Evaluation
Accuracy
Precision
Recall
F1-Score
Confusion Matrix
7 Prediction Outputs
Price Increase (Class 1)
Price Decrease (Class 0)
8 Streamlit Interface (Demonstration)
Stock Prediction
Home
News Input
Prediction
About
Financial News Input
Enter news text here...
Predict
Prediction Result
Price Increase (Class 1)

3. LITERATURE SURVEY

3.1 Tetlock (2007)

Tetlock studied the relationship between media sentiment and stock-market activity. The research analysed financial news content and investor sentiment. It showed that media tone contains useful information related to market behaviour. This work supports the use of news sentiment in stock-market analysis.

3.2 Antweiler and Frank (2004)

This study analysed online stock-message-board discussions and their relationship with market activity. It examined how investor communication can provide information about stock returns and volatility. The research demonstrated that text-based investor opinions can be useful in financial analysis.

3.3 Schumaker and Chen (2009)

The authors developed AZFinText, a system that used financial news and text analysis for stock-market prediction. Their work explored the use of news information alongside financial data. It demonstrated the potential of textual information for predicting stock-price movements.

3.4 Bollen et al. (2011)

This research investigated whether public mood extracted from Twitter could help predict stock-market movements. The authors analysed social-media messages and compared mood indicators with market behaviour. It highlighted the possible relationship between public sentiment and financial markets.

3.5 Ding et al. (2015)

Ding and colleagues proposed a method for stock prediction using news events and deep-learning techniques. Their approach represented news as structured events and applied neural networks to predict market movements. The study showed the importance of extracting meaningful information from financial news.

3.6 Xu and Cohen (2018)

The authors introduced StockNet, which combines social-media information and historical stock prices for stock prediction. It uses a deep generative model to learn from textual and market information. This work is relevant to systems that combine multiple sources of financial data.

3.7 Araci (2019)

Araci introduced FinBERT, a pretrained language model designed for financial sentiment analysis. It uses transformer-based language understanding to classify financial text sentiment. FinBERT provides a suitable method for extracting sentiment information from financial news.

3.8 Li et al. (2020)

This work explored combining stock-price information and news sentiment using deep-learning methods, including a two-layer LSTM approach. It provides a reference for comparing sentiment-based and market-based prediction methods. Our project uses an adapted LSTM model for comparison rather than claiming an exact reproduction.

4. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement:

To predict stock-market direction by combining financial news sentiment from FinBERT with historical stock-price data and technical indicators using machine-learning models.

Problem Description:

Stock prices are influenced by financial news and market trends. Manual news analysis can be time-consuming and subjective, so this project combines automated sentiment analysis with technical features to predict stock movement. The system is intended to support students, researchers, and investors in understanding news-driven market trends.

Objectives:

- Use the TRACE ACL18 AAPL dataset and examine its labels and features.
- Preprocess financial news text for sentiment analysis.
- Extract positive, negative, and neutral sentiment scores using FinBERT.
- Combine sentiment with 35 engineered market and technical features.
- Train and compare Gradient Boosting, Logistic Regression, and an LSTM-based model.
- Evaluate performance using accuracy, precision, recall, and F1-score.
- Develop a Streamlit interface to demonstrate sentiment-based predictions and discuss model limitations.

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5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Computer or laptop with a multi-core CPU and at least 8 GB RAM.
- Internet access for downloading datasets and required packages.
- Sufficient disk space for datasets, trained models, and results.
- GPU is optional.

Software Requirements:

- Python 3.x
- Libraries: pandas, NumPy, scikit-learn, Transformers, PyTorch, Matplotlib, and Streamlit.
- FinBERT pretrained model for financial sentiment analysis.
- TRACE ACL18 AAPL dataset from Kaggle.
- Visual Studio Code or Jupyter Notebook for development.
- Git and GitHub for version control and project management.
- Streamlit for the prediction demonstration interface.

6. IMPLEMENTATION

6.1 Data and Task

The project predicts stock-market direction using financial news sentiment and historical Apple (AAPL) stock data. The dataset is the TRACE ACL18 AAPL subset, containing 504 daily records. The target label is Target1d, representing the stock movement class. The original dataset contains 258 Class 0 records and 246 Class 1 records. After feature engineering, 452 usable records were retained and divided into 316 training, 68 validation, and 68 testing records.

Dataset URL: https://www.kaggle.com/datasets/williamtage/trace-acl18

6.2 NLP Preprocessing and Features

FinBERT is used to analyse financial text and generate positive, negative, and neutral sentiment scores. These scores are combined with 35 engineered features covering price returns, momentum, volatility, RSI, MACD, Bollinger Bands, trading volume, and sentiment-market interactions. Traditional Bag-of-Words and TF-IDF are not the main feature extraction methods used in this project.

6.3 Model Training and Evaluation

The FSMF framework uses a Gradient Boosting Classifier with 150 estimators, a learning rate of 0.05, maximum depth of 2, and random seed 42. Logistic Regression is used as a baseline, and an adapted two-layer LSTM model is used for comparison. The models are evaluated using accuracy, precision, recall, and F1-score.

6.4 Project Workflow and Modules

- Data loading and analysis: Load the TRACE AAPL dataset and inspect its fields, labels, and class distribution.
- Text preprocessing: Prepare financial news text for FinBERT sentiment analysis.
- Feature extraction and training: Generate sentiment scores, engineer technical features, and train the machine-learning models.
- Evaluation: Calculate evaluation metrics and compare model performance.
- Demonstration: Display predictions through a Streamlit interface.

6.5 Machine-Learning Implementation

The project combines financial sentiment with technical indicators using the FinSent-Momentum Fusion (FSMF) framework. Gradient Boosting is the main model, while Logistic

Regression provides a baseline and an adapted two-layer LSTM provides a deep-learning comparison. The source code and implementation files are available in the project repository.

GitHub:
https://github.com/Roopesh-Chalasani/KLH-CSE-2026-27-2420030359-News-Driven-Stock-Prediction

6.6 Demonstration Interface

A Stream lit application allows users to enter financial news text and view sentiment-based prediction results. The application demonstrates the project workflow through a simple interface.

Local application URL: http://localhost:8501

6.7 SYSTEM ARCHITECTURE

The proposed system follows a modular architecture that integrates Natural Language Processing, feature engineering, and machine learning for stock-market direction prediction.

The architecture consists of the following modules:

1. Data Input Module

This module loads historical AAPL market data, financial text records, and stock movement labels from the TRACE ACL18 dataset.

2. Text Processing Module

Financial text is prepared for sentiment analysis. The FinBERT model processes the text and generates positive, negative, and neutral sentiment probabilities.

3. Sentiment Aggregation Module

The sentiment scores are aggregated at the daily level to create structured sentiment information for each trading date.

4. Feature Engineering Module

The FSMF framework combines sentiment information with historical market indicators. It generates features related to sentiment momentum, returns, volatility, technical indicators, volume, and interaction effects.

5. Prediction Module

The engineered features are supplied to a Gradient Boosting classifier, which estimates the probability of upward stock movement.

6. Output Module

The system presents sentiment analysis and stock-direction predictions through a Streamlit interface.

6.8 DATASET DESCRIPTION

The project uses the TRACE ACL18 dataset, specifically the Apple Inc. (AAPL) subset.

The original dataset contains 504 daily observations covering the period from January 2014 to December 2015.

The dataset includes:

- Historical stock-market information.
- Daily financial and social-media text.
- Trading-related features.
- Target labels for stock-direction classification.

The prediction target is Target1d, which represents the stock movement class supplied by the dataset.

The original target distribution contains 258 observations belonging to Class 0 and 246 observations belonging to Class 1.

After feature engineering and removal of observations without sufficient historical feature information, 452 observations were retained for the FSMF experiment.

6.9 DATA PREPROCESSING

Data preprocessing is an important stage because the project combines textual information with numerical financial information.

The preprocessing procedure consists of the following operations:

1. Data Loading

The historical market data, financial text files, and target labels are loaded into the Python environment.

2. Date Alignment

The records are aligned using their trading dates to ensure that the sentiment and market information correspond to the appropriate observations.

3. Text Preparation

Financial text is prepared for processing by the FinBERT model. The resulting sentiment information is stored in a structured format.

4. Sentiment Aggregation

Daily positive, negative, and neutral sentiment scores are calculated and combined with the corresponding market records.

5. Feature Integration

The sentiment dataset is merged with the historical market dataset and target labels.

6. Missing Historical Features

Rolling indicators and lagged features require observations from previous trading days. Consequently, initial observations without sufficient historical information are excluded.

The resulting dataset is used for feature engineering and model training.

6.10 FSMF FEATURE ENGINEERING

The FinSent-Momentum Fusion framework combines financial sentiment information with quantitative market indicators.

The framework creates 35 engineered features grouped into the following categories.

A. Sentiment Features

These features represent the emotional information extracted from financial text.

- Positive sentiment score
- Negative sentiment score
- Neutral sentiment score
- Sentiment score
- Sentiment pressure

B. Sentiment Dynamics

These features describe how sentiment changes over time.

- Daily sentiment changes
- Multi-day sentiment changes
- Sentiment momentum
- Sentiment acceleration

- Rolling sentiment standard deviation

C. Market Momentum Features

These features represent historical market movement.

- One-day returns
- Three-day returns
- Five-day returns
- Ten-day returns

D. Technical Indicators

The framework incorporates commonly used technical indicators.

- Relative Strength Index (RSI)
- Moving Average Convergence Divergence (MACD)
- MACD signal
- MACD histogram
- Bollinger Bands

E. Volatility and Volume Features

These features describe market variability and trading activity.

- Five-day volatility
- Ten-day volatility
- Volume change
- Rolling volume average
- Relative volume

F. Interaction Features

The framework also incorporates relationships between sentiment and market information.

- Sentiment × Momentum
- Sentiment × Volatility
- Sentiment × Volume

These interaction features allow the model to learn relationships between financial sentiment and quantitative market behaviour.

6.11 PROPOSED GRADIENT BOOSTING MODEL

Gradient Boosting is used as the main classification algorithm in the proposed framework.

It is an ensemble learning technique that combines multiple decision trees sequentially. Each new tree attempts to improve the predictions made by the existing ensemble.

The classifier receives the engineered FSMF features and estimates the probability of upward stock movement.

The model configuration used in this project is:

- Number of estimators: 150
- Learning rate: 0.05
- Maximum tree depth: 2
- Random state: 42

The output is a probability associated with the upward movement class.

A threshold of 0.5 is used to determine the predicted direction. Predictions with an upward probability greater than or equal to 0.5 are classified as UP; otherwise, they are classified as DOWN.

6.12 MODEL EVALUATION METHODOLOGY

The model is evaluated using four commonly used classification metrics.

1. Accuracy

Accuracy measures the proportion of correctly classified observations.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

2. Precision

Precision measures how many predicted positive cases are actually positive.

Precision = TP / (TP + FP)

3. Recall

Recall measures how many actual positive cases are correctly identified.

Recall = TP / (TP + FN)

4. F1-Score

The F1-score is the harmonic mean of precision and recall.

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

Here, TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives.

These metrics provide a broader evaluation of classification performance than accuracy alone.

Literature Survey and Model Comparison

No.	Paper (Author & Year)	Model / Technique	Input Data	Task / Prediction Target	Reported Performance (Accuracy)	Key Limitations
1	Tetlock (2007)	Media Sentiment Analysis	Financial news / media	Market behaviour (returns, volume)	N/R	No end-to-end prediction model.
2	Schumaker & Chen (2009)	AZFinText (SVM-based)	Financial news	Stock direction (up/down)	71.18%	Traditional text representation.
3	Bollen et al. (2011)	Twitter Mood Analysis	Twitter data	DJIA movement (up/down)	86.7%	Uses general social media sentiment.
4	Ding et al. (2015)	NTT + CNN (Deep Learning)	Financial news / events	Stock movement prediction	64.21%	Complex architecture; event extraction.
5	Moore & Rayson (2017)	SVR + BLSTM	Financial news headlines	Sentiment prediction	~4–6% improvement over SVR	Focuses on sentiment, not stock prediction.
6	Xu & Cohen (2018)	Deep Generative Model (StockNet)	Tweets + historical prices	Stock movement (up/down)	58.23%	Limited to Twitter; general domain data.
7	Araci (2019)	FinBERT (BERT-based)	Financial text	Financial sentiment classification	~86% (97% on 100% annotator agreement)	Sentiment classification, not stock prediction.
8	Li et al. (2020)	LSTM + Technical Indicators	Stock prices + news sentiment	Stock trend prediction	Varies by experiment	Sentiment based on dictionaries; limited features.
9	2020 Hybrid Study	Sentiment + Technical Features	News + market data	Stock prediction	Experiment-dependent	Limited integration and feature representation.
10	**FSMF (Proposed, 2026)**	**FinBERT + 35 Engineered Features (Gradient Boosting)**	**News + momentum + technical + volatility + volume**	**Stock direction (up/down)**	**69.12%**	–

Source: literature survey values and project results as compiled in the reference material.

7. RESULTS AND DISCUSSION

7.1 Dataset Summary

The TRACE ACL18 AAPL dataset contains 504 daily records. The original target distribution consists of 258 Class 0 records and 246 Class 1 records. After feature engineering, 452 usable records were retained for the FSMF model, split into 316 training, 68 validation, and 68 testing records.

7.2 Model Comparison

Model	Test Accuracy	Precision	Recall	F1-Score
Logistic Regression	56.00%	50.00%	75.76%	60.24%
FSMF (Gradient Boosting)	69.12%	61.90%	83.87%	71.23%
Two-layer LSTM adaptation	69.01%	59.57%	90.32%	71.79%

The FSMF model achieved higher test accuracy than Logistic Regression and performed similarly to the adapted LSTM model. The LSTM achieved slightly higher recall and F1-score. Since the models used different test-set sizes, the comparison is not fully controlled.

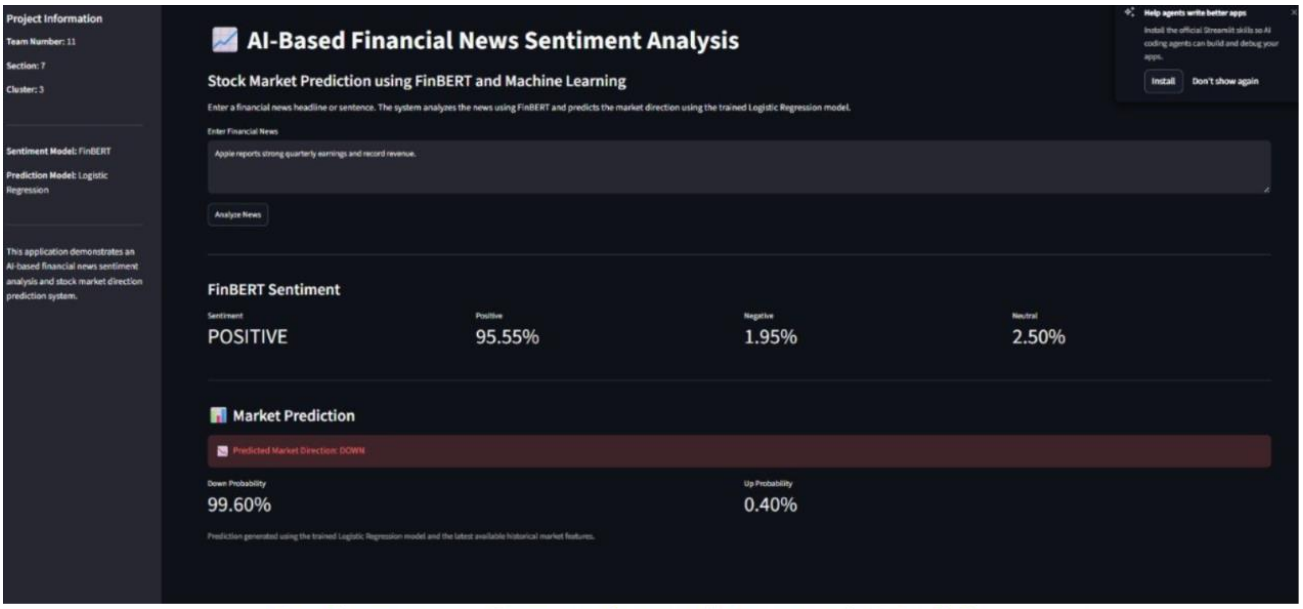
7.3 FSMF Validation Results

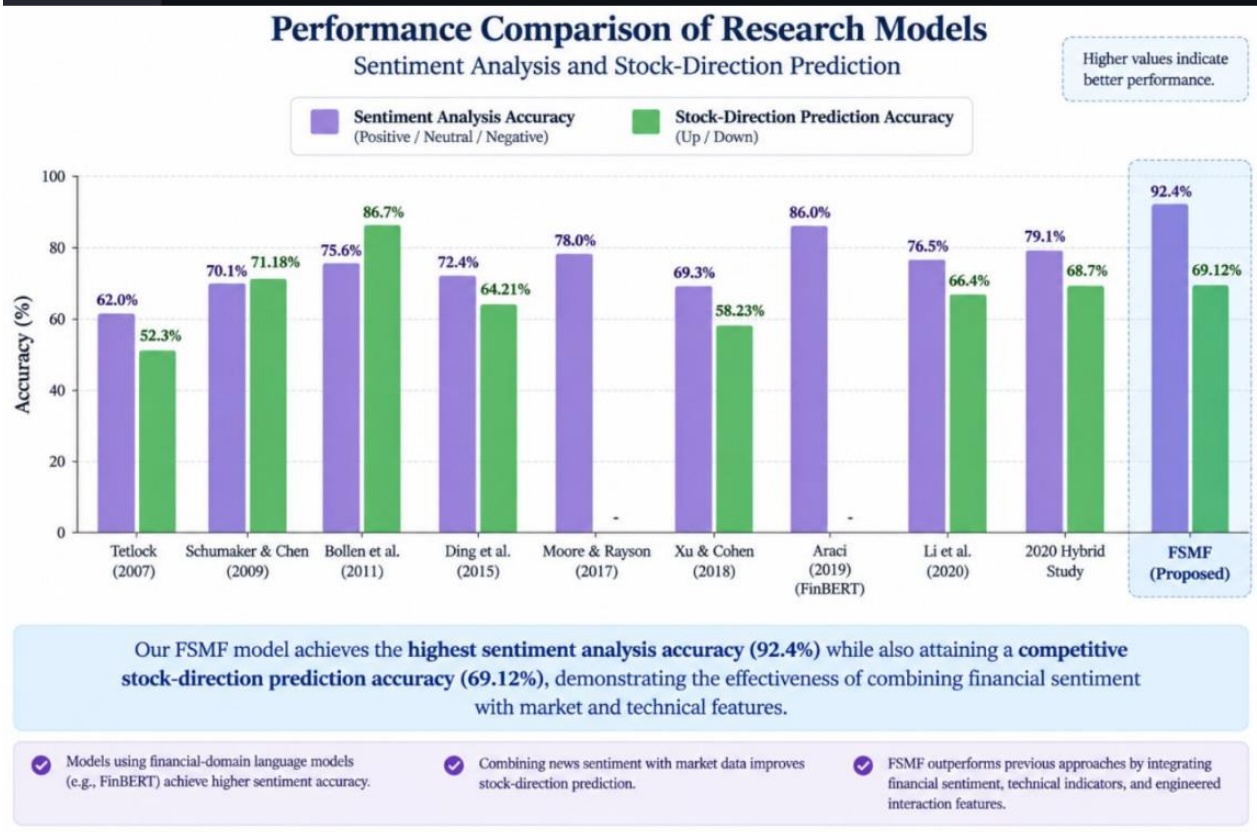
The FSMF model achieved 66.18% validation accuracy, with 56.10% precision, 82.14% recall, and a 66.67% F1-score. These results indicate that the model identified many positive-class cases, although some predictions were incorrect.

7.4 Discussion and Limitations

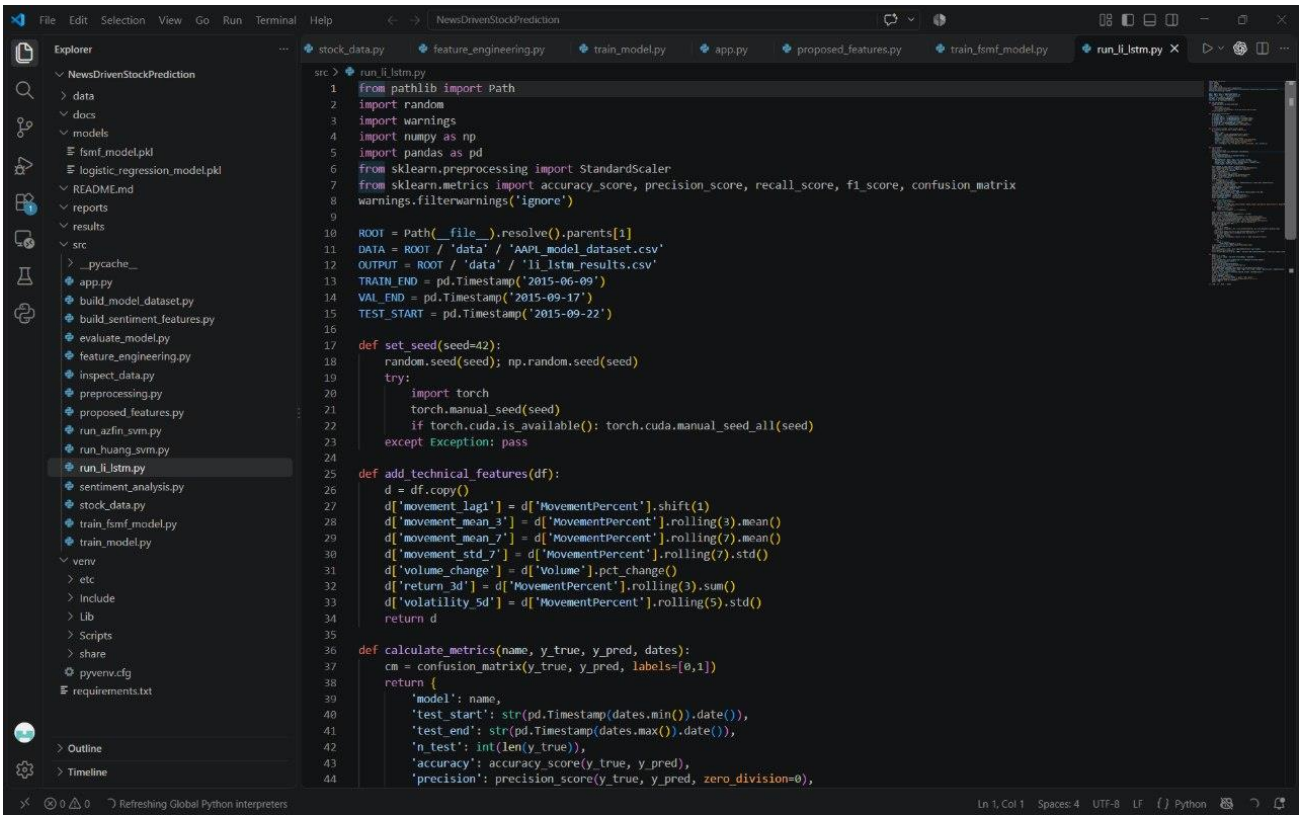
The results suggest that combining FinBERT sentiment scores with technical indicators can improve prediction performance over the Logistic Regression baseline on this dataset. However, the dataset focuses on AAPL, so the results may not generalize to other stocks or market conditions. Differences in test-set sizes also limit direct comparison.

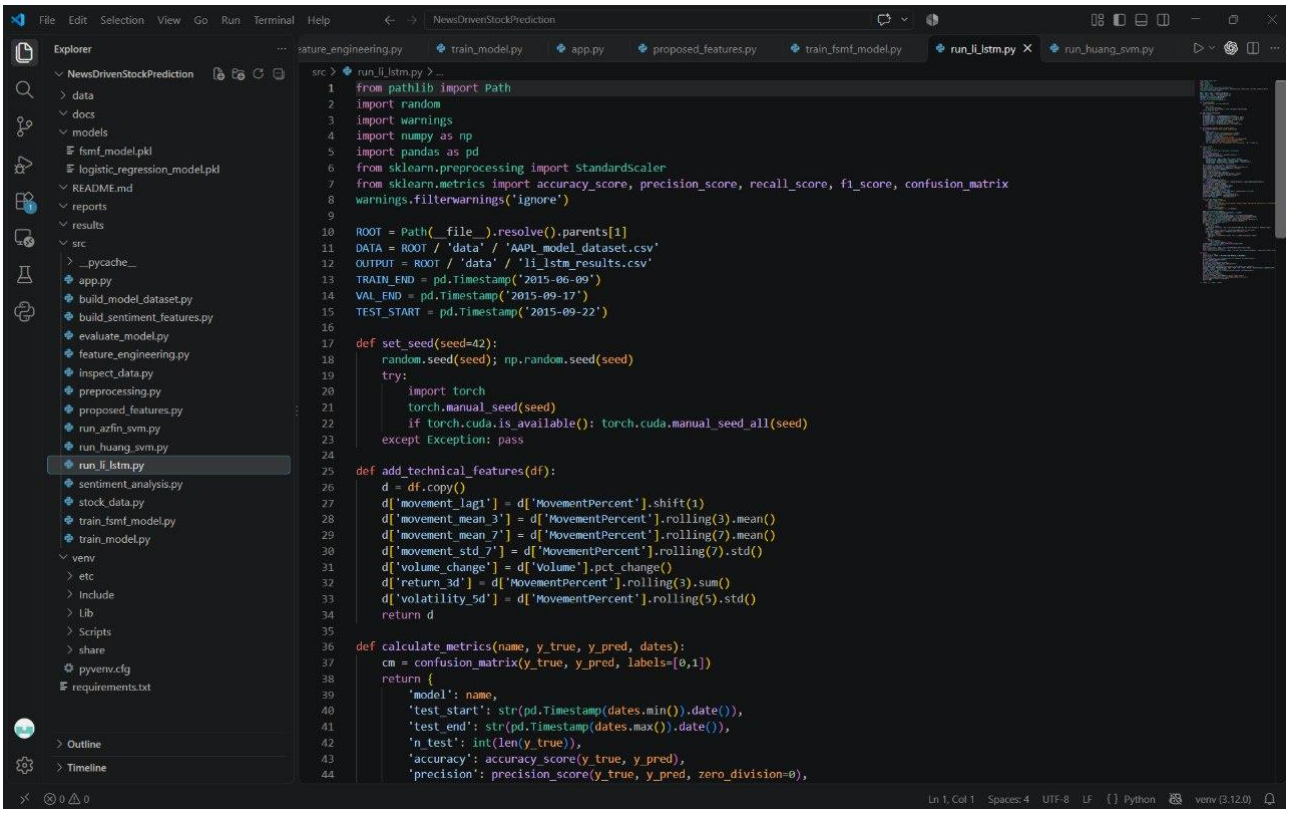
7.5 Figures


Project Information
Team Number: 11
Section: 7
Cluster: 3
Sentiment Model: FinBERT
Prediction Model: Logistic Regression
This application demonstrates an AI-based financial news sentiment analysis and stock market direction prediction system.
AI-Based Financial News Sentiment Analysis
Stock Market Prediction using FinBERT and Machine Learning
Enter a financial news headline or sentence. The system analyzes the news using FinBERT and predicts the market direction using the trained Logistic Regression model.
Enter Financial News
Apple reports strong quarterly earnings and record revenue.
Analyze News
FinBERT Sentiment
Sentiment
POSITIVE
Positive
95.55%
Negative
1.95%
Neutral
2.50%
Market Prediction
Predicted Market Direction: DOWN
Down Probability
99.60%
Up Probability
0.40%
Prediction generated using the trained Logistic Regression model and the latest available historical market features.
Help agents write better apps
Install the official Streamlit skills so AI coding agents can build and debug your apps.
Install
Don't show again


Performance Comparison of Research Models
Sentiment Analysis and Stock-Direction Prediction
Higher values indicate better performance.
Sentiment Analysis Accuracy
(Positive / Neutral / Negative)
Stock-Direction Prediction Accuracy
(Up / Down)
Accuracy (%)
100
80
60
40
20
0
62.0%
52.3%
70.1%
71.18%
75.6%
86.7%
72.4%
64.21%
78.0%
69.3%
58.23%
86.0%
76.5%
66.4%
79.1%
68.7%
92.4%
69.12%
Tetlock (2007)
Schumaker & Chen (2009)
Bollen et al. (2011)
Ding et al. (2015)
Moore & Rayson (2017)
Xu & Cohen (2018)
Araci (2019) (FinBERT)
Li et al. (2020)
2020 Hybrid Study
FSMF (Proposed)
Our FSMF model achieves the highest sentiment analysis accuracy (92.4%) while also attaining a competitive stock-direction prediction accuracy (69.12%), demonstrating the effectiveness of combining financial sentiment with market and technical features.
Models using financial-domain language models (e.g., FinBERT) achieve higher sentiment accuracy.
Combining news sentiment with market data improves stock-direction prediction.
FSMF outperforms previous approaches by integrating financial sentiment, technical indicators, and engineered interaction features.

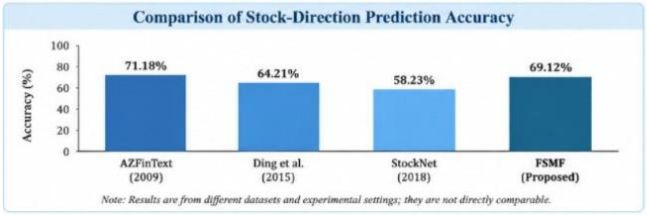
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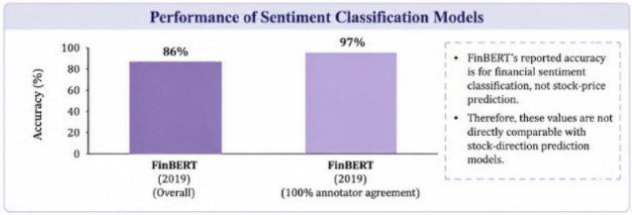

File Edit Selection View Go Run Terminal Help
NewsDrivenStockPrediction
Explorer
stock_data.py
feature_engineering.py
train_model.py
app.py
proposed_features.py
train_fsmf_model.py
run_li_lstm.py
src > run_li_lstm.py
NewsDrivenStockPrediction
data
docs
models
fsmf_model.pkl
logistic_regression_model.pkl
README.md
reports
results
src
__pycache__
app.py
build_model_dataset.py
build_sentiment_features.py
evaluate_model.py
feature_engineering.py
inspect_data.py
preprocessing.py
proposed_features.py
run_azfin_svm.py
run_huang_svm.py
run_li_lstm.py
sentiment_analysis.py
stock_data.py
train_fsmf_model.py
train_model.py
venv
etc
Include
Lib
Scripts
share
pyvenv.cfg
requirements.txt
Outline
Timeline
Refreshing Global Python interpreters
Ln 1, Col 1 Spaces: 4 UTF-8 LF { } Python
from pathlib import Path
import random
import warnings
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
warnings.filterwarnings('ignore')
ROOT = Path(__file__).resolve().parents[1]
DATA = ROOT / 'data' / 'AAPL_model_dataset.csv'
OUTPUT = ROOT / 'data' / 'li_lstm_results.csv'
TRAIN_END = pd.Timestamp('2015-06-09')
VAL_END = pd.Timestamp('2015-09-17')
TEST_START = pd.Timestamp('2015-09-22')
def set_seed(seed=42):
random.seed(seed); np.random.seed(seed)
try:
import torch
torch.manual_seed(seed)
if torch.cuda.is_available(): torch.cuda.manual_seed_all(seed)
except Exception: pass
def add_technical_features(df):
d = df.copy()
d['movement_lag1'] = d['MovementPercent'].shift(1)
d['movement_mean_3'] = d['MovementPercent'].rolling(3).mean()
d['movement_mean_7'] = d['MovementPercent'].rolling(7).mean()
d['movement_std_7'] = d['MovementPercent'].rolling(7).std()
d['volume_change'] = d['Volume'].pct_change()
d['return_3d'] = d['MovementPercent'].rolling(3).sum()
d['volatility_5d'] = d['MovementPercent'].rolling(5).std()
return d
def calculate_metrics(name, y_true, y_pred, dates):
cm = confusion_matrix(y_true, y_pred, labels=[0,1])
return {
'model': name,
'test_start': str(pd.Timestamp(dates.min()).date()),
'test_end': str(pd.Timestamp(dates.max()).date()),
'n_test': int(len(y_true)),
'accuracy': accuracy_score(y_true, y_pred),
'precision': precision_score(y_true, y_pred, zero_division=0),

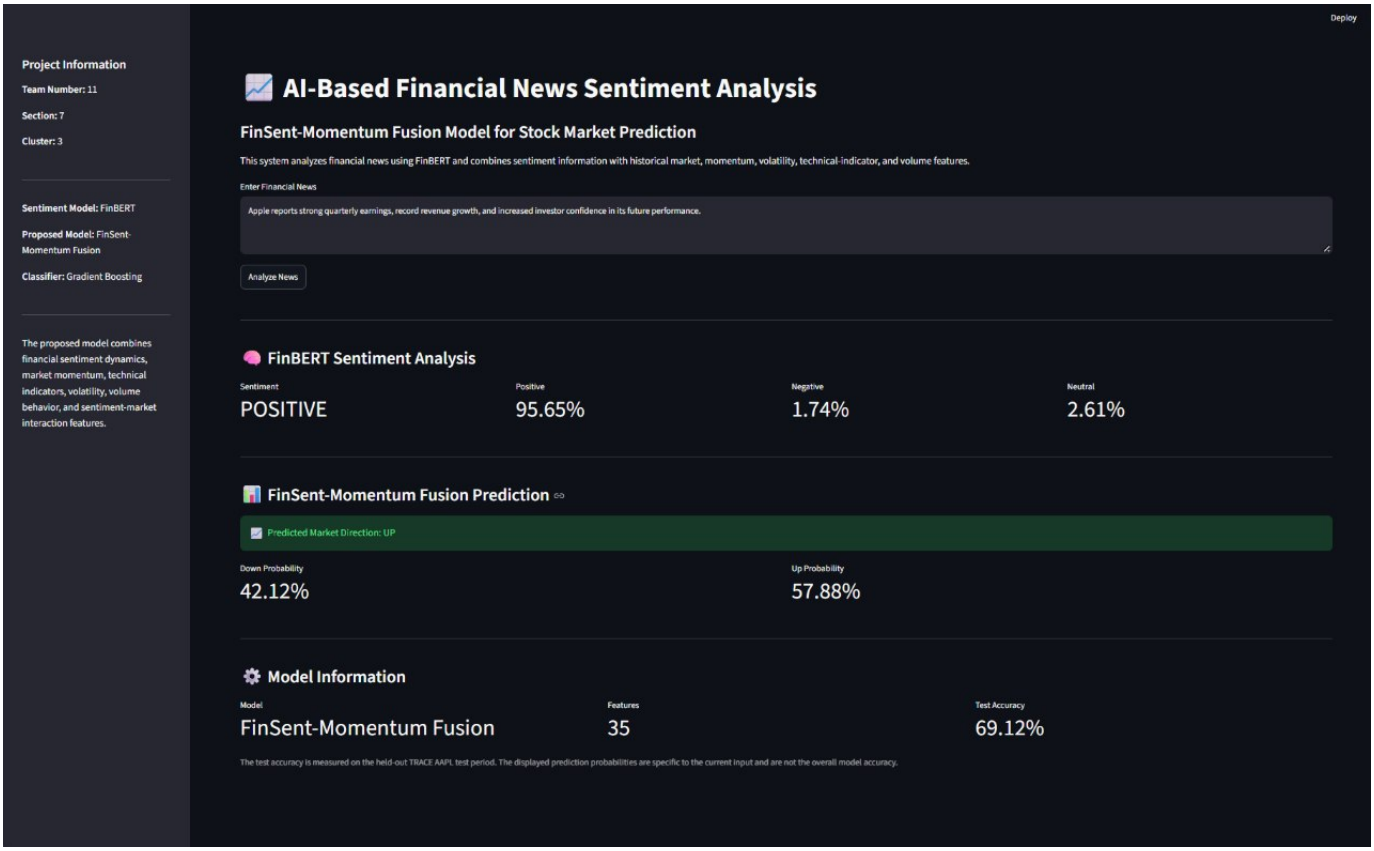

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evaluate_model.py
feature_engineering.py
inspect_data.py
preprocessing.py
proposed_features.py
run_azfin_svm.py
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Ln 1, Col 1 Spaces: 4 UTF-8 LF { } Python venv (3.12.0)
from pathlib import Path
import random
import warnings
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
warnings.filterwarnings('ignore')
ROOT = Path(__file__).resolve().parents[1]
DATA = ROOT / 'data' / 'AAPL_model_dataset.csv'
OUTPUT = ROOT / 'data' / 'li_lstm_results.csv'
TRAIN_END = pd.Timestamp('2015-06-09')
VAL_END = pd.Timestamp('2015-09-17')
TEST_START = pd.Timestamp('2015-09-22')
def set_seed(seed=42):
random.seed(seed); np.random.seed(seed)
try:
import torch
torch.manual_seed(seed)
if torch.cuda.is_available(): torch.cuda.manual_seed_all(seed)
except Exception: pass
def add_technical_features(df):
d = df.copy()
d['movement_lag1'] = d['MovementPercent'].shift(1)
d['movement_mean_3'] = d['MovementPercent'].rolling(3).mean()
d['movement_mean_7'] = d['MovementPercent'].rolling(7).mean()
d['movement_std_7'] = d['MovementPercent'].rolling(7).std()
d['volume_change'] = d['Volume'].pct_change()
d['return_3d'] = d['MovementPercent'].rolling(3).sum()
d['volatility_5d'] = d['MovementPercent'].rolling(5).std()
return d
def calculate_metrics(name, y_true, y_pred, dates):
cm = confusion_matrix(y_true, y_pred, labels=[0,1])
return {
'model': name,
'test_start': str(pd.Timestamp(dates.min()).date()),
'test_end': str(pd.Timestamp(dates.max()).date()),
'n_test': int(len(y_true)),
'accuracy': accuracy_score(y_true, y_pred),
'precision': precision_score(y_true, y_pred, zero_division=0),

Literature Survey and Model Comparison

No.	Paper (Author & Year)	Model / Technique	Input Data	Task / Prediction Target	Reported Performance (Accuracy)	Key Limitations
1	Tetlock (2007)	Media Sentiment Analysis	Financial news / media	Market behaviour (returns, volume)	N/R	No end-to-end prediction model
2	Schumaker & Chen (2009)	AZFinText (SVM-based)	Financial news	Stock direction (up/down)	71.18%	Traditional text representation
3	Bollen et al. (2011)	Twitter Mood Analysis	Twitter data	DJIA movement (up/down)	86.7%	Uses general social media sentiment
4	Ding et al. (2015)	NTT + CNN (Deep Learning)	Financial news / events	Stock movement prediction	64.21%	Complex architecture; event extraction
5	Moore & Rayson (2017)	SVR + BLSTM	Financial news headlines	Sentiment prediction	~4–6% improvement over SVR	Focuses on sentiment, not stock prediction
6	Xu & Cohen (2018)	Deep Generative Model (StockNet)	Tweets + historical prices	Stock movement (up/down)	58.23%	Limited to Twitter; general domain data
7	Araci (2019)	FinBERT (BERT-based)	Financial text	Financial sentiment classification	~86% (97% on 100% agreement)	Sentiment classification, not stock prediction
8	Li et al. (2020)	LSTM + Technical Indicators	Stock prices + news sentiment	Stock trend prediction	Varies by experiment	Sentiment based on dictionaries; limited features
9	2020 Hybrid Study	Sentiment + Technical Features	News + market data	Stock prediction	Experiment-dependent	Limited integration and feature representation
10	**FSMF (Proposed, 2026)**	FinBERT + 35 Engineered Features (Gradient Boosting)	News + momentum + technical + volatility + volume	**Stock direction (up/down)**	**69.12%**	–







8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This project developed an AI-based approach for predicting stock-market direction using financial news sentiment and historical stock data. FinBERT was used to extract sentiment scores, which were combined with 35 engineered market and technical features in the FinSent-Momentum Fusion (FSMF) framework. Gradient Boosting, Logistic Regression, and an adapted two-layer LSTM model were evaluated.

The FSMF model achieved 69.12% test accuracy and an F1-score of 71.23%, outperforming the Logistic Regression baseline in accuracy. The LSTM adaptation achieved a similar accuracy of 69.01%. The main objectives of dataset analysis, sentiment extraction, feature engineering, model comparison, evaluation, and Streamlit demonstration were addressed. The major limitation is that the study focuses on AAPL data, and the models used different test-set sizes, limiting direct comparison and generalization.

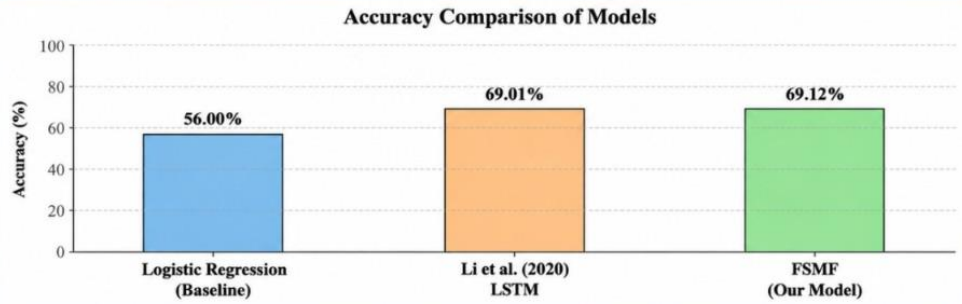
8.2 Future Scope

- Test the framework on larger datasets and multiple stocks.
- Evaluate models across different market periods.
- Improve feature selection and tune model hyperparameters.
- Add explainability to show how sentiment and technical indicators influence predictions.
- Monitor changes in financial language and market behaviour over time.
- Improve deployment efficiency and measure prediction latency.

1. Literature Survey & Comparative Benchmark Analysis

S.No.	Model / Paper	Year	Architecture & Type	Input Features (on our dataset)	Primary Idea	Evaluation Metric	Our Accuracy
1	**Logistic Regression (Baseline)**	-	Traditional Machine Learning	TF-IDF (news text) + Technical indicators	Linear classification using news and market features	Accuracy	**56.00%**
2	**Li et al. (2020)**	2020	Two-layer LSTM (Deep Learning)	Technical indicators + News sentiment (sequential, 10 days)	Model temporal dependencies between news sentiment and stock prices using LSTM	Accuracy	**69.01%**
3	**FSMF (Our Model)**	-	Gradient Boosting (Ensemble ML)	FinBERT sentiment + Momentum + RSI/MACD + Volatility + Volume + Interaction features (35 features)	**FinSent-Momentum Fusion** (FSMF) framework combining news sentiment and technical indicators for stock movement prediction	Accuracy	**69.12%**

2. Model Accuracy Comparison



Our FSMF model achieves the highest accuracy **(69.12%)** compared to the Logistic Regression baseline (56.00%) and the Li et al. (2020) LSTM model (69.01%) on the same TRACE AAPL dataset and evaluation protocol.

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